

Exercise Sheet 5 — Solutions

Exercise 1 – Structure factor of an infinite linear atomic chain

We consider a 1D crystal along the x -axis. The *direct lattice* has primitive vector $\vec{a} = a\hat{x}$. The *reciprocal lattice* is 1D with primitive vector $\vec{b}$ satisfying $\vec{a} \cdot \vec{b} = 2\pi$, hence:

$$\vec{b} = \frac{2\pi}{a} \hat{x}.$$

Reciprocal lattice points are at $G_n = n\vec{b} = \frac{2\pi n}{a}$, $n \in \mathbb{Z}$.

The *structure factor* for a Bragg peak at reciprocal lattice vector $\vec{G}$ is

$$S_{\vec{G}} = \sum_j f_j e^{i\vec{G} \cdot \vec{r}_j},$$

where the sum runs over all atoms in the unit cell (basis), f_j is the atomic form factor of atom j , and $\vec{r}_j$ its position within the unit cell.

Assume $f \propto Z$ (atomic number), so $f_{\text{Cu}} \approx 29$, $f_{\text{Zn}} \approx 30$, $f_{\text{Au}} \approx 79$.

(a) Uniform chain of Cu atoms

Lattice: primitive, 1 atom per unit cell at $\vec{r}_1 = 0$.

Unit cell: $[0, a)$, atom at $x = 0$.

Reciprocal lattice: Points at

$$G_n = \frac{2\pi n}{a}, \quad n \in \mathbb{Z}.$$

Sketch: equally spaced points on a line, spacing $2\pi/a$.

Structure factor:

$$S_{G_n} = f_{\text{Cu}} e^{iG_n \cdot 0} = f_{\text{Cu}}.$$

All Bragg peaks have equal intensity: $I_n \propto |S_{G_n}|^2 = f_{\text{Cu}}^2$.

(b) Chain with additional atom at center

Now the unit cell is doubled: atoms at $x = 0$ and $x = a/2$. To keep the lattice primitive, we can either: - Keep primitive vector a , basis: two atoms at 0 and $a/2$, or - Use new primitive vector $a' = a/2$, basis: one atom.

We use the *conventional* description with lattice constant a and two-atom basis (more intuitive for diffraction).

Lattice: still periodic with period a , but unit cell now contains two atoms at:

$$\vec{r}_1 = 0, \quad \vec{r}_2 = \frac{a}{2} \hat{x}.$$

Reciprocal lattice: unchanged — still $G_n = \frac{2\pi n}{a}$, $n \in \mathbb{Z}$, because the periodicity is still a .

However, the *structure factor* now filters which G_n give nonzero intensity.

Sketch: same as (a), but now we expect systematic absences.

(c) Structure factor for (a) and (b)

Already computed: $S_{G_n} = f_{\text{Cu}}$ for all n .

Let atom 1 be Cu ($f_1 = f_{\text{Cu}}$), atom 2 be element X with $f_2 = f_X$ (could be Cu, Zn, or Au). Then:

$$S_{G_n} = f_1 e^{iG_n \cdot 0} + f_2 e^{iG_n \cdot (a/2)} = f_1 + f_2 e^{i \frac{2\pi n}{a} \cdot \frac{a}{2}} = f_1 + f_2 e^{i\pi n}.$$

Since $e^{i\pi n} = (-1)^n$, we obtain:

$$S_{G_n} = \begin{cases} f_1 + f_2 & \text{if } n \text{ even,} \\ f_1 - f_2 & \text{if } n \text{ odd.} \end{cases}$$

Special cases:

- $f_1 = f_2 = f_{\text{Cu}}$: then $S_{G_n} = 2f_{\text{Cu}}$ for n even, 0 for n odd.
So all *odd-order* reflections vanish (e.g. $n = \pm 1, \pm 3, \dots$). This is the classic case of a *centrosymmetric diatomic chain* with equal atoms — analogous to bcc in 3D (where $(h+k+l)$ odd vanish).
- $f_1 = f_{\text{Cu}}, f_2 = f_{\text{Zn}} \approx f_{\text{Cu}}$: nearly equal, so odd peaks very weak but nonzero.
- $f_1 = f_{\text{Cu}}, f_2 = f_{\text{Au}} \gg f_{\text{Cu}}$: then $S_{G_n} \approx f_{\text{Au}}(1 + (-1)^n \frac{f_{\text{Cu}}}{f_{\text{Au}}})$, so even peaks $\approx 2f_{\text{Au}}$, odd peaks $\approx f_{\text{Au}} - f_{\text{Cu}} > 0$, both present, but intensity ratio $\sim (f_{\text{Au}} + f_{\text{Cu}})^2 / (f_{\text{Au}} - f_{\text{Cu}})^2$ large.

(d) Every 4th atom replaced

Now we have a *superlattice* with period $4a$. Primitive vector $\vec{a}' = 4a\hat{x}$. One unit cell contains 4 atoms:

$$\vec{r}_1 = 0, \quad \vec{r}_2 = a, \quad \vec{r}_3 = 2a, \quad \vec{r}_4 = 3a.$$

Atoms 1–3: Cu (f_{Cu}), atom 4: X (f_X).

Reciprocal lattice now has spacing $\frac{2\pi}{4a} = \frac{\pi}{2a}$, i.e. points at

$$G_m = \frac{2\pi m}{4a} = \frac{\pi m}{2a}, \quad m \in \mathbb{Z}.$$

Structure factor:

$$S_{G_m} = f_{\text{Cu}} (e^{iG_m \cdot 0} + e^{iG_m a} + e^{iG_m 2a}) + f_X e^{iG_m 3a}.$$

Compute phases:

$$G_m \cdot ka = \frac{\pi m}{2a} \cdot ka = \frac{\pi mk}{2}, \quad k = 0, 1, 2, 3.$$

So:

$$S_{G_m} = f_{\text{Cu}} (1 + e^{i\pi m/2} + e^{i\pi m}) + f_X e^{i3\pi m/2}.$$

Use $e^{i\pi m} = (-1)^m$, and note that $e^{i\pi m/2}$ cycles every 4:

$$e^{i\pi m/2} = \begin{cases} 1 & m \equiv 0 \pmod{4}, \\ i & m \equiv 1 \pmod{4}, \\ -1 & m \equiv 2 \pmod{4}, \\ -i & m \equiv 3 \pmod{4}. \end{cases}$$

Then compute S_{G_m} modulo 4:

$$m = 4\ell: e^{i\pi m/2} = 1, e^{i\pi m} = 1, e^{i3\pi m/2} = 1 \Rightarrow S = f_{\text{Cu}}(1 + 1 + 1) + f_X \cdot 1 = 3f_{\text{Cu}} + f_X.$$

$$m = 4\ell + 1: 1 + i + (-1) = i, e^{i3\pi/2} = -i \Rightarrow S = f_{\text{Cu}} \cdot i + f_X \cdot (-i) = i(f_{\text{Cu}} - f_X).$$

$$m = 4\ell + 2: 1 + (-1) + 1 = 1, e^{i3\pi} = -1 \Rightarrow S = f_{\text{Cu}} \cdot 1 + f_X \cdot (-1) = f_{\text{Cu}} - f_X.$$

$$m = 4\ell + 3: 1 + (-i) + (-1) = -i, e^{i9\pi/2} = e^{i\pi/2} = i \Rightarrow S = f_{\text{Cu}} \cdot (-i) + f_X \cdot i = i(f_X - f_{\text{Cu}}).$$

Thus:

$$S_{G_m} = \begin{cases} 3f_{\text{Cu}} + f_X & m \equiv 0 \pmod{4}, \\ i(f_{\text{Cu}} - f_X) & m \equiv 1 \pmod{4}, \\ f_{\text{Cu}} - f_X & m \equiv 2 \pmod{4}, \\ i(f_X - f_{\text{Cu}}) & m \equiv 3 \pmod{4}. \end{cases}$$

Consequences: - For $X = \text{Zn} \approx \text{Cu}$: $f_X \approx f_{\text{Cu}}$, so $S \approx 4f$ for $m \equiv 0$, and ≈ 0 otherwise — only every 4th peak survives (superlattice reflections at $G = \frac{2\pi}{4a} \cdot 4\ell = \frac{2\pi\ell}{a}$, i.e. the original Bragg peaks, but now with enhanced intensity and suppressed satellites). - For $X = \text{Au}$ ($f_X \gg f_{\text{Cu}}$): all peaks nonzero; relative intensities differ strongly.

Note: The original periodicity a is broken, so the reciprocal lattice is denser ($\pi/(2a)$ spacing), but only every 4th point has large intensity if $f_X \approx f_{\text{Cu}}$.

Exercise 2 – Diffraction from a simple cubic lattice with a two-atom basis

The crystal is simple cubic with lattice constant a , basis:

$$\vec{r}_1 = \delta a \hat{x}, \quad \vec{r}_2 = -\delta a \hat{x}, \quad 0 < \delta < \frac{1}{4}.$$

Reciprocal lattice: $\vec{G}_{hkl} = \frac{2\pi}{a}(h, k, l)$, $h, k, l \in \mathbb{Z}$.

Structure factor:

$$S_{hkl} = f_1 e^{i\vec{G}_{hkl} \cdot \vec{r}_1} + f_2 e^{i\vec{G}_{hkl} \cdot \vec{r}_2}.$$

Compute dot products:

$$\vec{G}_{hkl} \cdot \vec{r}_1 = \frac{2\pi}{a}(h, k, l) \cdot (\delta a, 0, 0) = 2\pi h \delta,$$

$$\vec{G}_{hkl} \cdot \vec{r}_2 = \frac{2\pi}{a}(h, k, l) \cdot (-\delta a, 0, 0) = -2\pi h \delta.$$

So:

$$S_{hkl} = f_1 e^{i2\pi h \delta} + f_2 e^{-i2\pi h \delta} = (f_1 + f_2) \cos(2\pi h \delta) + i(f_1 - f_2) \sin(2\pi h \delta).$$

(a) Identical atoms ($f_1 = f_2 = f$)

Then the imaginary part vanishes:

$$S_{hkl} = f (e^{i2\pi h\delta} + e^{-i2\pi h\delta}) = 2f \cos(2\pi h\delta).$$

Intensity:

$$I_{hkl} \propto |S_{hkl}|^2 = 4f^2 \cos^2(2\pi h\delta).$$

Since intensity depends only on h (not k, l), we have:

$$I_{hkl} = I_{h00}, \quad \text{and} \quad I_{\bar{h}kl} = I_{hkl} \quad (\text{since } \cos(-x) = \cos x).$$

So:

$$I_{hkl} = I_{\bar{h}kl} = 4f^2 \cos^2(2\pi h\delta).$$

Note: For $h = 0$, $I_{0kl} = 4f^2$ (maximal). For small δ , peaks decrease slowly with $|h|$.

(b) Different atoms ($f_1 \neq f_2$)

Now:

$$S_{hkl} = (f_1 + f_2) \cos(2\pi h\delta) + i(f_1 - f_2) \sin(2\pi h\delta).$$

Intensity:

$$|S_{hkl}|^2 = (f_1 + f_2)^2 \cos^2(2\pi h\delta) + (f_1 - f_2)^2 \sin^2(2\pi h\delta).$$

Using $\cos^2 = \frac{1}{2}(1 + \cos 4\pi h\delta)$, $\sin^2 = \frac{1}{2}(1 - \cos 4\pi h\delta)$, or directly:

$$|S_{hkl}|^2 = f_1^2 + f_2^2 + 2f_1 f_2 \cos(4\pi h\delta).$$

(Verify by expansion: $(A \cos \theta + iB \sin \theta)(A \cos \theta - iB \sin \theta) = A^2 \cos^2 \theta + B^2 \sin^2 \theta$, with $A = f_1 + f_2$, $B = f_1 - f_2$.)

Vanishing reflections: $I_{hkl} = 0$ iff $|S_{hkl}| = 0$, i.e.

$$(f_1 + f_2) \cos(2\pi h\delta) = 0 \quad \text{and} \quad (f_1 - f_2) \sin(2\pi h\delta) = 0.$$

Since $f_1, f_2 > 0$, $f_1 + f_2 \neq 0$. So need:

$$\cos(2\pi h\delta) = 0 \quad \Rightarrow \quad 2\pi h\delta = \frac{\pi}{2} + n\pi \Rightarrow h\delta = \frac{1}{4} + \frac{n}{2},$$

and simultaneously,

$$\sin(2\pi h\delta) = \pm 1 \neq 0,$$

so the second condition requires $f_1 - f_2 = 0$. Thus, for $f_1 \neq f_2$, $|S_{hkl}| > 0$ for all h, k, l — no *complete* extinction.

However, if $\delta = \frac{1}{4}$, then $2\pi h\delta = \frac{\pi h}{2}$, and:

$$S_{hkl} = \begin{cases} f_1 + f_2 & h \equiv 0 \pmod{4}, \\ i(f_1 - f_2) & h \equiv 1 \pmod{4}, \\ -(f_1 + f_2) & h \equiv 2 \pmod{4}, \\ -i(f_1 - f_2) & h \equiv 3 \pmod{4}. \end{cases}$$

So for $f_1 = f_2$, odd h give $S = 0$ — see part (c).

(c) Identical atoms, $\delta = \frac{1}{4}$

Then:

$$S_{hkl} = 2f \cos\left(2\pi h \cdot \frac{1}{4}\right) = 2f \cos\left(\frac{\pi h}{2}\right).$$

Now:

$$\cos\left(\frac{\pi h}{2}\right) = \begin{cases} 1 & h \equiv 0 \pmod{4}, \\ 0 & h \equiv 1, 3 \pmod{4}, \\ -1 & h \equiv 2 \pmod{4}. \end{cases}$$

Hence:

$$S_{hkl} = \begin{cases} 2f & h \text{ divisible by } 4, \\ 0 & h \text{ odd}, \\ -2f & h \equiv 2 \pmod{4}. \end{cases}$$

So for odd h , the structure factor vanishes: $S_{hkl} = 0$. Physically, the two atoms are at $x = \pm a/4$, i.e. separated by $a/2$ — the unit cell is effectively doubled, and odd- h reflections vanish, analogous to a bcc-like 1D superlattice.

Exercise 3 – Cubic perovskite BaTiO₃

The conventional unit cell is *simple cubic* with side a . Atom positions:

Ba: corners $(0, 0, 0)$ — but each corner atom is shared by 8 cells, so contribution: 1 atom per cell.

Ti: body center $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ — fully inside $\rightarrow$ 1 atom.

O: face centers — $(\frac{1}{2}, \frac{1}{2}, 0)$, $(\frac{1}{2}, 0, \frac{1}{2})$, $(0, \frac{1}{2}, \frac{1}{2})$ — each shared by 2 cells, $3 \times \frac{1}{2} = \frac{3}{2}$? Wait — no: 6 faces, but only 3 unique positions in the basis (the other 3 are translations by lattice vectors).

Correct basis:

Standard description: one formula unit BaTiO₃ per primitive cell? Actually, for the simple cubic setting, the unit cell contains:

$$\text{Ba} : 8 \times \frac{1}{8} = 1, \quad \text{Ti} : 1, \quad \text{O} : 6 \times \frac{1}{2} = 3.$$

So 5 atoms per cubic unit cell $\rightarrow$ correct.

(a) Lattice and basis

Lattice: simple cubic, primitive vectors $\vec{a}_1 = a\hat{x}$, $\vec{a}_2 = a\hat{y}$, $\vec{a}_3 = a\hat{z}$.

Basis (5 atoms):

$$\begin{aligned} \vec{r}_{\text{Ba}} &= (0, 0, 0), \\ \vec{r}_{\text{Ti}} &= \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) a, \\ \vec{r}_{\text{O}_1} &= \left(\frac{1}{2}, \frac{1}{2}, 0\right) a, \\ \vec{r}_{\text{O}_2} &= \left(\frac{1}{2}, 0, \frac{1}{2}\right) a, \\ \vec{r}_{\text{O}_3} &= \left(0, \frac{1}{2}, \frac{1}{2}\right) a. \end{aligned}$$

This is the standard perovskite description.

(b) Intensity of first four Bragg reflections

Reciprocal lattice: $\vec{G}_{hkl} = \frac{2\pi}{a}(h, k, l)$.

Structure factor:

$$S_{hkl} = f_{\text{Ba}} e^{i\vec{G} \cdot \vec{r}_{\text{Ba}}} + f_{\text{Ti}} e^{i\vec{G} \cdot \vec{r}_{\text{Ti}}} + f_{\text{O}} \left[e^{i\vec{G} \cdot \vec{r}_{\text{O}_1}} + e^{i\vec{G} \cdot \vec{r}_{\text{O}_2}} + e^{i\vec{G} \cdot \vec{r}_{\text{O}_3}} \right].$$

Compute phases (in units of 2π):

$$\vec{G} \cdot \vec{r} / (2\pi) = hx + ky + lz,$$

so:

$$\begin{aligned} \vec{G} \cdot \vec{r}_{\text{Ba}} / 2\pi &= 0, \\ \vec{G} \cdot \vec{r}_{\text{Ti}} / 2\pi &= \frac{1}{2}(h + k + l), \\ \vec{G} \cdot \vec{r}_{\text{O}_1} / 2\pi &= \frac{1}{2}(h + k), \\ \vec{G} \cdot \vec{r}_{\text{O}_2} / 2\pi &= \frac{1}{2}(h + l), \\ \vec{G} \cdot \vec{r}_{\text{O}_3} / 2\pi &= \frac{1}{2}(k + l). \end{aligned}$$

Thus:

$$S_{hkl} = f_{\text{Ba}} + f_{\text{Ti}} e^{i\pi(h+k+l)} + f_{\text{O}} \left[e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)} \right].$$

Using $e^{i\pi n} = (-1)^n$, define $\varepsilon_h = (-1)^h$, etc. Then:

$$S_{hkl} = f_{\text{Ba}} + f_{\text{Ti}} \varepsilon_h \varepsilon_k \varepsilon_l + f_{\text{O}} (\varepsilon_h \varepsilon_k + \varepsilon_h \varepsilon_l + \varepsilon_k \varepsilon_l).$$

Now compute for low-index reflections. The *first four* (by increasing $|G| \propto \sqrt{h^2 + k^2 + l^2}$) are:

$$(100), (110), (111), (200).$$

Compute each:

1. $(hkl) = (100)$: $\varepsilon_h = -1$, $\varepsilon_k = \varepsilon_l = 1$.

$$S_{100} = f_{\text{Ba}} + f_{\text{Ti}}(-1)(1)(1) + f_{\text{O}} [(-1)(1) + (-1)(1) + (1)(1)] = f_{\text{Ba}} - f_{\text{Ti}} + f_{\text{O}}(-1-1+1) = f_{\text{Ba}} - f_{\text{Ti}} - f_{\text{O}}.$$

2. (110) : $\varepsilon_h = \varepsilon_k = -1$, $\varepsilon_l = 1$.

$$S_{110} = f_{\text{Ba}} + f_{\text{Ti}}(-1)(-1)(1) + f_{\text{O}} [(-1)(-1) + (-1)(1) + (-1)(1)] = f_{\text{Ba}} + f_{\text{Ti}} + f_{\text{O}}(1-1-1) = f_{\text{Ba}} + f_{\text{Ti}} - f_{\text{O}}.$$

3. (111): $\varepsilon_h = \varepsilon_k = \varepsilon_l = -1$.

$$S_{111} = f_{\text{Ba}} + f_{\text{Ti}}(-1)^3 + f_{\text{O}} [(-1)^2 + (-1)^2 + (-1)^2] = f_{\text{Ba}} - f_{\text{Ti}} + 3f_{\text{O}}.$$

4. (200): $\varepsilon_h = 1, \varepsilon_k = \varepsilon_l = 1$ (since $(-1)^2 = 1$).

$$S_{200} = f_{\text{Ba}} + f_{\text{Ti}}(1) + f_{\text{O}}(1 + 1 + 1) = f_{\text{Ba}} + f_{\text{Ti}} + 3f_{\text{O}}.$$

Now use $f_{\text{Ba}} = 7f_{\text{O}}, f_{\text{Ti}} = 3f_{\text{O}}$. Set $f_{\text{O}} = f$ for simplicity:

$$S_{100} = 7f - 3f - f = 3f \quad \Rightarrow \quad I_{100} \propto 9f^2,$$

$$S_{110} = 7f + 3f - f = 9f \quad \Rightarrow \quad I_{110} \propto 81f^2,$$

$$S_{111} = 7f - 3f + 3f = 7f \quad \Rightarrow \quad I_{111} \propto 49f^2,$$

$$S_{200} = 7f + 3f + 3f = 13f \quad \Rightarrow \quad I_{200} \propto 169f^2.$$

Intensity ratios (normalized to I_{100}):

$$I_{100} : I_{110} : I_{111} : I_{200} = 9 : 81 : 49 : 169 = 1 : 9 : \frac{49}{9} \approx 5.44 : \frac{169}{9} \approx 18.78.$$

Or as integers (multiply by 9):

$$\boxed{I_{100} : I_{110} : I_{111} : I_{200} = 9 : 81 : 49 : 169.}$$

Note: The (100) reflection is allowed (unlike in fcc), because the basis breaks the fcc symmetry — BaTiO₃ in cubic phase is actually *primitive cubic* with a basis, not fcc.